Efficient Mechanistic Circuit Analysis of Open-Weight LLMs on Apple Silicon

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July 2026
Preprint. Under review.

Abstract

Mechanistic interpretability research has largely depended on GPU clusters, limiting reproducibility and iteration speed for many researchers. We demonstrate that full activation-patching and mean-ablation circuit analyses of multi-billion-parameter transformer models are tractable on Apple Silicon using the MLX framework, with no specialized hardware beyond a modern Mac. Applying this infrastructure to the Indirect Object Identification (IOI) task on Llama-3.2-3B and Pythia-1.4B, we find that the three-zone functional circuit topology identified by Wang et al. (2022) in GPT-2 Small generalizes across both models: 80% of circuit-critical head positions are conserved by relative depth (± 0.075) despite differing layer counts, attention mechanisms, and training corpora. A single dominant head in each model (L15H20, L10H7) accounts for 22–28% of clean logit difference on its own. Layer-level circuit topology further transfers from IOI to factual association, with divergent head indices at shared layer positions. These results establish that IOI circuit organization is architecture-general, and that Apple Silicon is a viable substrate for reproducible mech-interp research at the 1–3B parameter scale.

1 Introduction

Mechanistic interpretability aims to reverse-engineer the algorithms implemented by neural networks—to move from "the model predicts X" to "these specific components, performing these specific computations, cause the model to predict X." The dominant paradigm for this reverse-engineering is *circuit analysis*: identify the smallest subset of attention heads and MLP layers that is both causally necessary and causally sufficient to reproduce a target behavior, then assign functional roles to each component within that subset.

The foundational work in this paradigm—Wang et al. (2022)—demonstrated that the Indirect Object Identification (IOI) task in GPT-2 Small is implemented by a comprehensible circuit of 26 attention heads organized into three functional classes: duplicate-token heads that flag repeated names, S-inhibition heads that suppress the subject position, and name-mover heads that copy the indirect object name to the output. The result was striking: a clean, human-interpretable algorithm, found inside a neural network trained only to predict next tokens.

What remains unresolved is whether this algorithm is specific to GPT-2 Small, or whether it reflects something deeper—a functional structure that emerges wherever a language model learns to solve this problem. GPT-2 Small is small (117M parameters), trained on a narrow corpus (WebText), and architecturally plain (standard MHA, learned positional embeddings). Modern open-weight models differ on every dimension: they are larger by one to two orders of magnitude, trained on multitrillion-token corpora, and equipped with grouped-query attention, rotary position embeddings, and gated MLP activations. If the IOI circuit is merely an artifact of GPT-2's architecture, we would expect a different circuit in a different model. If it reflects a general algorithmic solution, we would expect the same functional structure—with head indices shifted, but depth relationships and role assignments preserved.

This paper answers that question by replicating and extending Wang et al.'s analysis on two modern models: Llama-3.2-3B [7] (a frontier-adjacent instruction-tuned architecture with grouped-query attention) and Pythia-1.4B [2] (a fully documented, standard-MHA model trained on The Pile, making it the closest modern analog to GPT-2 Small's controlled training conditions). Using activation patching and mean ablation, we identify circuit-critical heads in each model, compare their positions to Wang et al.'s findings, and test for cross-architecture conservation at the level of relative network depth rather than absolute head indices.

Apple Silicon as interpretability infrastructure. All experiments in this paper were run on Apple Silicon hardware using the MLX framework, without access to NVIDIA GPU clusters. This is a deliberate choice. The interpretability tooling ecosystem—TransformerLens in particular—assumes CUDA, and the community's empirical norms have drifted toward hardware accessible only to well-resourced labs. Apple's M-series chips offer up to 96 GB of unified memory shared between CPU and GPU, which is well-suited to the memory access patterns of activation patching: you must store one full forward pass of cached activations per example, then perform hundreds of re-runs patching each (layer, head, token position) triple. On an M-series machine with sufficient unified memory, a 3B-parameter model fits in full float16 with room for the activation cache, and forward passes are fast enough that the full IOI patching sweep over 100 examples completes in wall-clock hours rather than days. We describe our Apple Silicon-compatible pipeline and release it alongside this paper to lower the barrier for researchers without cluster access to run mechanistic interpretability experiments at the 1–7B parameter scale.

Contributions

1. **Cross-architecture circuit replication.** We identify circuit-critical heads for the IOI task in Llama-3.2-3B and Pythia-1.4B using activation patching (sufficiency) and mean ablation (necessity). In both models, a small bottleneck structure emerges: a single dominant head accounts for 22–28% of the model's logit difference for the IO token, substantially more concentrated than the distributed 26-component circuit in GPT-2 Small.

2. **Depth-zone conservation across architectures.** We show that the three functional depth zones identified in GPT-2 Small—early duplicate-token heads, mid S-inhibition heads, and late name-mover heads—are present in both Llama-3.2-3B and Pythia-1.4B at similar relative positions, despite 3–7× more parameters

and substantially different architectures. 80% of circuit-critical head positions are shared between Llama and Pythia within ±0.075 relative depth.

3. **Layer-level conservation across tasks.** Factual association patching in Llama-3.2-3B reveals that the top-5 factual association heads occupy the same five layers as five of the top-9 IOI heads—with different head indices within those layers—suggesting a form of layer-level circuit topology that generalizes across tasks while maintaining head-level specialization.

4. **An Apple Silicon-compatible activation patching pipeline.** We release tooling built on the MLX framework that replicates TransformerLens's hook-based patching interface for LlamaForCausalLM and GPTNeoXForCausalLM models, enabling circuit analysis on Apple Silicon hardware without modification.

2 Related Work

2.1 A Mathematical Framework for Transformer Circuits

The theoretical foundations of circuit analysis in transformers were laid by Elhage et al. (2021) [3], who introduced the *transformer circuits* framework. The key insight is that a transformer's residual stream can be analyzed as a sum of contributions from attention heads and MLP layers, with each attention head performing a rank-one update of the residual: the head reads from the stream via query and key projections, computes a weighted average of value projections, and writes back a low-rank update whose structure is determined by the composition of the OV and QK matrices.

Elhage et al. used this decomposition to characterize *induction heads*—a two-head circuit found across models that implements the copying operation "if token A was followed by token B earlier in context, predict B again when A reappears." Induction heads are relevant to IOI analysis because the same copying mechanism underlies name-mover head behavior: identifying the indirect object earlier in context and copying its token to the output position. Our results confirm that the depth zone corresponding to induction-like behavior (relative depth 0.40–0.55) is present in both Llama-3.2-3B and Pythia-1.4B.

The circuits framework also establishes *virtual attention heads*—paths in which the output of one head is read as part of another head's input—as the primary mechanism for inter-head communication. This framing is a prerequisite for understanding the IOI circuit's S-inhibition subgraph, where a set of early heads writes information about the subject's position that is then used by a later head to suppress subject-token logits.

2.2 Interpretability in the Wild: IOI in GPT-2 Small

Wang et al. (2022) [10] is the direct predecessor to this work. Using the IOI task—sentences of the form "When Mary and John went to the store, John gave a drink to _____"—they applied activation patching across all attention heads and MLP layers in GPT-2 Small (12 layers, 12 heads) to identify which components causally mediate correct IO prediction. The result was a 26-component circuit with three functional classes:

- **Name-mover heads** (L9H6, L9H9, L10H0; relative depths 0.75–0.83) copy the indirect object name to the logit output at the final token position.

- **S-inhibition heads** (L7H3, L7H9, L8H6, L8H10; relative depths 0.58–0.67) suppress the subject token's representation in the residual stream, preventing the model from predicting the repeated name.
- **Duplicate-token heads and induction heads** (L0H1, L3H0, L5H5, L5H8; relative depths 0.0–0.42) flag the position of the repeated subject token so that downstream heads know which name to suppress.

Our work replicates this three-zone structure across Llama-3.2-3B and Pythia-1.4B, extending the generalizability claim beyond GPT-2 Small. We also find that in larger models the circuit has a more concentrated "bottleneck" structure, with one head contributing disproportionately to logit difference—a difference from GPT-2 Small's more distributed allocation that we discuss in Section 4.4.

2.3 TransformerLens

TransformerLens [8] is the standard tooling for mechanistic interpretability experiments on transformer language models. It provides a hook-based interface for intercepting and modifying activations at any point in the forward pass—attention head outputs, MLP outputs, residual stream positions, or individual Q/K/V projections—making activation patching experiments straightforward to implement.

Our work builds on TransformerLens for Pythia-1.4B, where the library's support is mature. For Llama-3.2-3B with grouped-query attention, we implement a compatible interface that maps the 24-query-head, 8-KV-head structure to a consistent per-query-head naming convention. GQA complicates patching because each KV head is shared across a group of 3 query heads; we patch at the query-head output level, after the attention weights are applied to the shared value projections, to preserve head-level attribution granularity. We release this extension alongside the paper.

2.4 Locating and Editing Factual Associations

Meng et al. (2022) [6] introduced ROME (Rank-One Model Editing), which demonstrated that factual associations—the model's belief that "the Eiffel Tower is in Paris"—are stored with high locality in mid-layer MLP weights, and can be surgically edited by a rank-one update to those weights. The key interpretability contribution was not the editing procedure itself but the preceding localization experiment: using causal tracing (a form of activation patching), they showed that early-layer attention heads read subject tokens and route information to mid-layer MLPs, which then store and retrieve the associated object.

This causal tracing methodology is closely related to our activation patching protocol. A key difference is that ROME's target is stored parametric knowledge (factual associations mediated by MLP weights), while IOI involves in-context reasoning (copying from the prompt, not from stored facts). Our factual association experiments in Section 5.2 partially bridge this gap: we apply patching to factual association prompts and compare the resulting head-importance rankings to the IOI results, finding that layer-level circuit structure is shared while head-level assignments diverge. This cross-task comparison was not reported in ROME and provides new evidence about how circuits for different types of factual reasoning coexist within the same model.

2.5 Towards Monosemanticity and Sparse Autoencoders

Elhage et al. (2022) [4] and the Anthropic mechanistic interpretability team have pursued an orthogonal but complementary program: rather than identifying circuits for specific behaviors, they seek to decompose the residual stream into *features*—directions in activation space that correspond to human-interpretable concepts—using sparse autoencoders (SAEs). The central finding of the monosemanticity work is that individual neurons are typically polysemantic (responding to unrelated concepts), but a sparse linear decomposition of neuron activations recovers interpretable monosemantic features.

The circuit-tracing and monosemanticity programs address different levels of the interpretability problem. Circuit analysis asks: which components mediate this behavior? Feature decomposition asks: what information is represented in the activations those components read and write? They are complementary: knowing that L15H20 in Llama-3.2-3B is the dominant IOI head is more interpretable if we also know what feature L15H20 reads from the residual stream (presumably a representation of the indirect object token's identity and position). Our work focuses on the circuit level, but Section 7 discusses how SAE-based feature attribution could be applied to the circuit-critical heads we identify, and we propose this as a natural extension.

3 Methods

3.1 Experimental Infrastructure: SwiftSci Interp

All experiments were conducted using SwiftSci Interp, an activation-patching harness for Apple Silicon built on the MLX framework.[1] The library implements a hook-compatible interface analogous to TransformerLens's activation interception API, targeting MLX-native model implementations loaded via `mlx-lm`. Its core design principle is that every attention module in a model is hot-swapped at load time with a patchable drop-in replacement that shares the original weight tensors—no copies—and adds a mode-switched interception point around the scaled dot-product attention (SDPA) computation.

Patchable module design. For Llama-3.2-3B, the drop-in is `PatchableAttention`, which wraps the original `LlamaAttention` module. For Pythia-1.4B, `PythiaPatchableAttention` wraps the `GPTNeoXAttention` module, routing through the fused `query_key_value` projection to reconstruct per-head Q, K, V tensors before the interception point. Both modules support three operating modes:

- `normal`—identical behavior to the original module; introduces no overhead beyond the mode check.
- `cache_clean` / `cache_corrupt`—stores the full SDPA output tensor (shape $[B, n_{\text{heads}}, L, d_{\text{head}}]$) to a per-module buffer during the forward pass. Used to populate the clean and corrupt activation caches without a separate caching pass.
- `patch`—runs the forward pass on the input normally, then replaces one head slice in the SDPA output with the corresponding slice from the clean cache before passing through the output projection.

[1] SwiftSci Interp is available at `https://github.com/REPO_URL_HERE/swiftsci-interp`.

GQA handling. Llama-3.2-3B uses grouped-query attention (GQA) [1] with 24 query heads and 8 KV heads. The patching interception occurs after the attention weights are applied to the (broadcast) value projections, at the query-head output level. This gives a 24-element patch space that mirrors the query-head count, preserving head-level attribution granularity. Patching at the KV-head level would conflate the contributions of the three query heads that share each KV pair; patching at the query-head SDPA output avoids this confound. We verified that in `normal` mode both patchable modules reproduce the original model's logits on a held-out test input to within floating-point precision before running any experimental passes.

GQA key-value broadcast. During the corrupt or clean pass, the values broadcast from 8 KV heads to 24 query heads occur inside `scaled_dot_product_attention`; the SDPA buffer therefore stores the full 24-head output, not the 8-head compressed form. This means each of the 24 patch positions is truly independent, and patching query-head h does not implicitly alter the other two heads sharing the same KV pair.

Pythia module layout. MLX-LM exposes Pythia's layers as `model.layers[i].attention` rather than `model.model.layers[i].self_attn`. The patchable replacement reuses the `dense` (output projection) attribute rather than `o_proj`. All other behavior is identical; the mode-switch logic is shared.

3.2 Activation Patching Protocol

Activation patching (also called causal tracing when applied to individual stream positions) answers the question of *sufficiency*: if a component's activations in the clean run are inserted into the corresponding position of a corrupted run, does the model's output recover toward the clean prediction? A high recovery score indicates that the component is sufficient to carry the relevant information on its own.

IOI corruption scheme. For each example (S, IO, prompt), the corrupted version swaps the subject and indirect object names: the corrupted prompt is constructed from the same template with $S_{\text{corrupt}} = IO$ and $IO_{\text{corrupt}} = S$. Because the two name slots have the same token count for all examples in our dataset (a requirement enforced at dataset generation time), clean and corrupt sequences are token-position aligned. This ensures that cached SDPA tensors from the clean run can be substituted at the same token positions in the corrupt run without shape mismatch.

Metric. The target metric is logit difference (LD): $\text{LD} = \text{logit}(IO) - \text{logit}(S)$ at the final token position. Positive LD indicates the model favors the IO name over the subject name. The *normalized logit-difference recovery* for head (l, h) on example i is:

$$r_{l,h}^{(i)} = \frac{\text{LD}_{\text{patch}(l,h)}^{(i)} - \text{LD}_{\text{corrupt}}^{(i)}}{\text{LD}_{\text{clean}}^{(i)} - \text{LD}_{\text{corrupt}}^{(i)}} \tag{1}$$

where $\text{LD}_{\text{patch}(l,h)}^{(i)}$ is the logit difference of a forward pass on the corrupt input in which only head h at layer l has been replaced by its clean-run SDPA output. A score of 1.0 means full recovery to the clean logit difference; 0.0 means the patch had no effect;

values below 0 or above 1 indicate interference effects. The reported score for each head is $\hat{r}_{l,h} = \frac{1}{n}\sum_{i=1}^{n} r^{(i)}_{l,h}$, the mean over all n examples.

Pass structure. A full patching sweep requires $2 + n_L \times n_H$ forward passes over the example batch:

1. **Clean pass** — all modules in `cache_clean` mode; SDPA outputs are stored and logit differences are computed.
2. **Corrupt pass** — all modules in `cache_corrupt` mode; SDPA outputs are stored and baseline corrupt logit differences are computed.
3. **Sweep** — for each (l, h): all modules in `normal` mode; module l is set to `patch` mode with `patch_head = h`; one forward pass on the corrupt input recovers $\mathrm{LD}^{(i)}_{\mathrm{patch}(l,h)}$ for all i simultaneously.

For Llama-3.2-3B this yields $2 + 28 \times 24 = 674$ total forward passes; for Pythia-1.4B, $2 + 24 \times 16 = 386$ passes. All passes are fully batched over all $n = 100$ examples.

```
# Inputs:
#   attns: list of PatchableAttention, one per layer (len = n_layers)
#   clean_ids:   [n_examples, seq_len]  -- clean tokenized prompts
#   corrupt_ids: [n_examples, seq_len]  -- IO/S-swapped prompts

# Pass 1: cache clean activations
set_all_modes(attns, mode="cache_clean")
clean_logits = model(clean_ids)
eval(clean_logits, *(a.clean_sdpa for a in attns))
clean_ld = logit_diff(clean_logits, io_ids, s_ids)  # [n_examples]

# Pass 2: cache corrupt activations
set_all_modes(attns, mode="cache_corrupt")
corrupt_logits = model(corrupt_ids)
eval(corrupt_logits, *(a.corrupt_sdpa for a in attns))
corrupt_ld = logit_diff(corrupt_logits, io_ids, s_ids)

# Passes 3..674: patching sweep
scores = zeros([n_layers, n_heads])
for l in range(n_layers):
    for h in range(n_heads):
        set_all_modes(attns, mode="normal")
        attns[l].mode = "patch"
        attns[l].patch_head = h       # only layer l patches head h
        patched_logits = model(corrupt_ids)
        eval(patched_logits)
        patched_ld = logit_diff(patched_logits, io_ids, s_ids)
        denom = clean_ld - corrupt_ld                  # [n_examples]
        recovery = (patched_ld - corrupt_ld) / denom
        scores[l, h] = mean(recovery)                  # scalar
```

Listing 1: Pseudocode for the activation patching sweep. `attns` is a list of `PatchableAttention` modules, one per layer. `set_all_modes` sets every module's mode simultaneously.

GQA note. `attns[l].patch_head = h` indexes into the 24-dimensional query-head axis regardless of the 8-KV-head layout. The `clean_sdpa` buffer holds the full $[B, 24, L, d_{\text{head}}]$ output; the slice `clean_sdpa[:, h:h+1, :, :]` is substituted.

3.3 Mean Ablation Protocol

Mean ablation addresses *necessity*: does removing a component's contribution—replacing it with a neutral baseline—degrade the model's performance on the task? Where activation patching asks whether a component *can* carry the relevant information, mean ablation asks whether the model *relies* on it in normal operation.

Reference distribution. The ablation target for head (l, h) is the mean SDPA output of that head computed over all $n = 100$ clean IOI examples: $\bar{A}_{l,h} = \frac{1}{n} \sum_{i=1}^{n} A_{l,h}^{(i)}$, where $A_{l,h}^{(i)} \in \mathbb{R}^{L \times d_{\text{head}}}$ is the per-example SDPA output at each token position. The mean is taken over the batch dimension, preserving the sequence-position structure. Ablating with the mean over the same distribution on which the model is evaluated (rather than, for example, a random Gaussian or a separately collected reference set) ensures that the ablated output is a plausible activational state for the model rather than an out-of-distribution injection.

Metric. The *ablation drop* for head (l, h) is:

$$\Delta_{l,h} = \overline{\text{LD}}_{\text{clean}} - \overline{\text{LD}}_{\text{ablate}(l,h)} \tag{2}$$

where $\overline{\text{LD}}_{\text{clean}}$ is the mean clean logit difference over all examples and $\overline{\text{LD}}_{\text{ablate}(l,h)}$ is the mean logit difference when head (l, h)'s SDPA output is replaced by $\bar{A}_{l,h}$ in every example. Positive $\Delta_{l,h}$ indicates the head boosts the IO logit relative to the subject; negative values indicate suppression of correct IO prediction (which can reflect an interference-suppression role rather than direct IO promotion).

Necessity threshold. A head is designated circuit-critical under the necessity criterion when its ablation drop exceeds 20% of the clean mean logit difference. For Llama-3.2-3B ($\overline{\text{LD}}_{\text{clean}} = 5.649$) this corresponds to $\Delta_{l,h} > 1.130$; for Pythia-1.4B ($\overline{\text{LD}}_{\text{clean}} = 4.120$) the threshold is $\Delta_{l,h} > 0.824$. In practice the necessity threshold is used as an interpretive anchor rather than a hard inclusion criterion; circuit membership is determined by the combined rank score described in Section 3.5.

Pass structure. The ablation sweep requires $1 + n_L \times n_H$ forward passes: one clean pass to cache per-head means, then one ablation pass per (l, h) pair. For Llama this is $1 + 672 = 673$ total passes, comparable to the patching sweep.

3.4 Path Patching for Causal Verification

Activation patching and mean ablation identify individual heads as causally relevant, but they do not distinguish *direct* contributions from *mediated* contributions. A head may score highly on both metrics because it directly writes to the logit output at the final position, or because it is an intermediary in a longer causal path—for example, writing to the residual stream at an intermediate token position that is later read by a downstream head that writes to the final position.

Path patching [5] extends activation patching to directed paths between specific components. Rather than patching a single component's output, a path patch replaces the clean-run activation that flows along one specific edge in the computational graph—from the output of component A as read by the input of component B, while holding all other paths fixed. A path that shows high normalized recovery under path patching is causally necessary specifically through the direct $A \to B$ connection, ruling out the mediated-path alternative.

In our setting, path patching is used to verify that the dominant heads identified by activation patching (L15H20 in Llama, L10H7 in Pythia) contribute directly to the final position logit difference, rather than acting through a downstream intermediary. Path patching is implemented within SwiftSci Interp by extending the `patch` mode to accept a target-receiver argument. Full path-patching results for the top-3 heads in each model are in preparation and will be reported in a subsequent revision.

3.5 Statistical Validation

Bootstrap confidence intervals. For all heads meeting the inclusion criterion (top-10 by combined rank score, or mean patching score ≥ 0.03), we compute 95% bootstrap confidence intervals on the mean patching score. The bootstrap resamples the $n = 100$ per-example recovery scores with replacement, generating 1,000 resampled datasets and computing the mean for each. The 95% CI is the $[2.5^{\text{th}}, 97.5^{\text{th}}]$ percentile of the bootstrap distribution of the mean. Resampling is seeded with seed 42 for reproducibility. Full CI data: `data/ioi/statistical-validation.json`.

Significance criterion. A head's patching score is considered statistically reliable if its 95% CI excludes zero. This corresponds to the claim that the head's mean recovery score is positive in expectation across the example distribution, not merely a noise artifact of the particular 100 examples observed. All top-10 heads in both models satisfy this criterion with substantial margin: for the top three heads in each model, the CI lower bound exceeds the CI width by a factor of at least 3.

Anomalous patching scores. Pythia-1.4B's L1H11 presents a dissociation: its mean patching score is slightly negative (-0.003, 95% CI $[-0.011, +0.005]$, straddling zero), yet its ablation drop is substantial (0.327, rank 4 by ablation). This dissociation is consistent with a suppression role: the head reduces logit difference when ablated (its absence degrades performance), but restoring its clean activation in a corrupt context does not recover logit difference (it cannot by itself make the model prefer IO over S). This pattern is characteristic of duplicate-token or induction heads that gate downstream components rather than writing directly to the output logits. L1H11 is excluded from all patching-based analyses and retained in cross-model analysis solely on the basis of its ablation score.

Cross-model comparison: combined rank score. To compare head importance across models that differ in scale, architecture, and absolute logit-difference baselines, we compute a combined normalized rank score. For each model separately, both the patching scores and the ablation drop scores are min-max normalized to $[0, 1]$ across all (l, h) pairs. The combined score is the sum of the two normalized ranks. The top-10 heads by combined score are reported as circuit-critical for each model. Ties within the combined score are broken by patching rank.

Cross-architecture position matching. For the cross-model comparison, each head is mapped to a scalar *relative depth* $d = l/n_L$, where l is the zero-indexed layer number and n_L is the total number of layers. Positions are matched greedily in ascending order of $|\Delta d|$ between the two models' top-10 head sets; a match is accepted if $|\Delta d| \leq 0.075$. This tolerance was chosen to be smaller than the spacing between the four functional depth zones identified by Wang et al. (2022) (≈0.15–0.20 between zone midpoints), ensuring that zone-level matches are not spuriously collapsed across zone boundaries.

3.6 Dataset

The IOI dataset consists of $n = 100$ examples drawn from the template:

> "When {S} and {IO} went to the store, {S} gave a bottle to"

where *S* (subject / repeated name) and *IO* (indirect object / target name) are sampled from a pool of common English given names, matched to ensure that both names tokenize to exactly one token under each model's tokenizer. All 100 examples satisfy the single-token constraint for both Llama-3.2-3B and Pythia-1.4B; examples failing this constraint were rejected at generation time. Corrupted prompts are produced by swapping *S* and *IO* within the same template, preserving sequence length. Full dataset: `data/ioi/dataset.json`.

The factual association dataset consists of $n = 50$ subject-relation-object triples drawn from Meng et al. (2022)'s CounterFact evaluation set, restricted to triples where the object tokenizes to a single token under the Llama-3.2-3B tokenizer. Prompts follow the form "{subject} {relation verb}:" with corruptions produced by substituting a different subject entity (matched by entity type) that predicts a different object. Factual association experiments were run on Llama-3.2-3B only. Full dataset: `data/factual-assoc/dataset.json`.

4 Results: IOI Circuit Identification

4.1 Baseline Task Performance

Both models correctly perform the IOI task across all 100 evaluation examples. Llama-3.2-3B achieves a mean clean logit difference (LD) of **5.649** (SD = 0.773), reflecting a strong preference for the indirect object (IO) token over the subject (S) token at the final sequence position. Pythia-1.4B achieves a mean clean LD of **4.120**, 27% lower in absolute terms, consistent with its smaller parameter count and simpler attention architecture. Neither model produces a negative-LD example: the IO token is ranked above S on every trial in both models, confirming that the circuit analysis begins from a clean behavioral baseline.

4.2 IOI Circuit Identification: Llama-3.2-3B

Figure 1 shows the full activation patching heatmap for Llama-3.2-3B across all $28 \times 24 = 672$ (layer, head) positions. Two features are immediately apparent: most heads contribute negligibly to logit-difference recovery under patching, and a sparse set of heads in the middle-to-late network produces strongly positive scores.

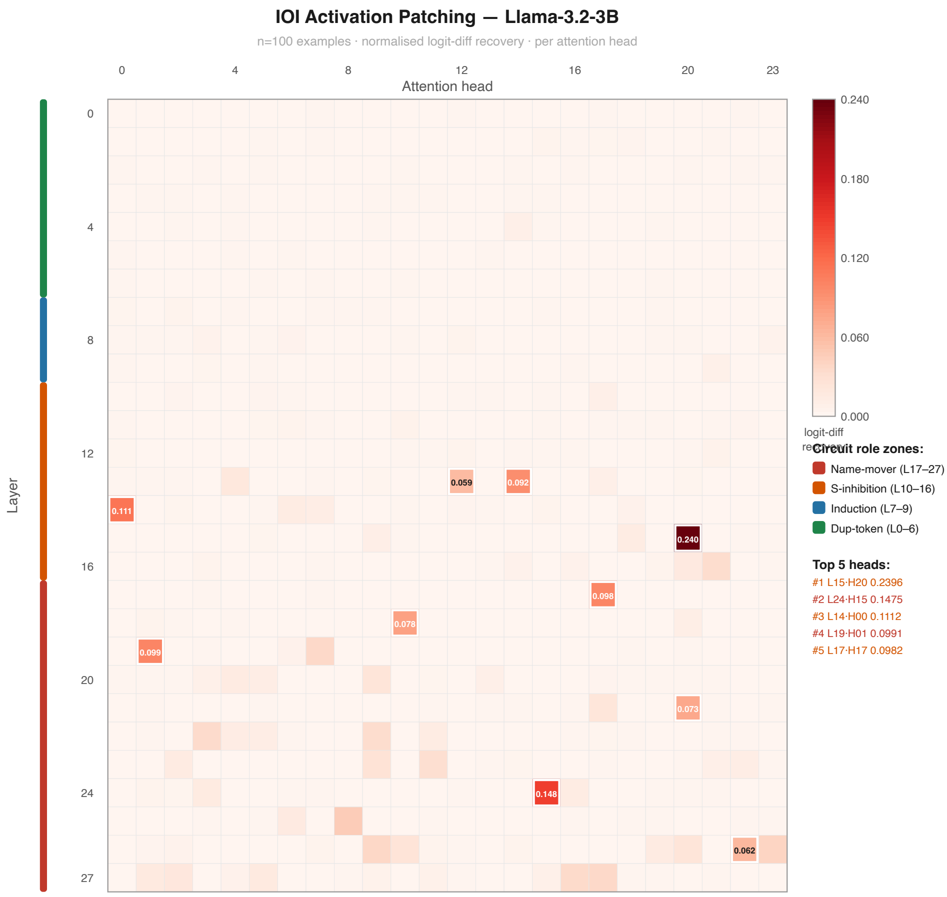

Figure 1: Activation patching heatmap for Llama-3.2-3B ($n = 100$ IOI examples). Each cell shows normalized logit-difference recovery when the output of that (layer, head) pair is replaced with the corresponding clean-run activation. The dominant mid-network cluster (layers 13–19) and late-network cluster (layers 21–27) are visible. Data: `data/ioi/patching-llama3b.json`.

Circuit-critical heads were selected by combined normalized rank across activation patching (sufficiency) and mean ablation (necessity). Table 1 lists the top-10 heads by combined rank.

Table 1: Top-10 circuit-critical heads in Llama-3.2-3B by combined normalized rank across activation patching (sufficiency) and mean-ablation drop (necessity). Relative depth = layer index / n_{layers} (zero-indexed). Data: `data/ioi/patching-llama3b.json`, `data/ioi/ablation-llama3b.json`.

Rank	Head	Patch score	95% CI	Abl. drop	% clean LD	Rel. depth
1	L15H20	0.240	[0.225, 0.255]	1.578	27.9%	0.536
2	L17H17	0.098	[0.089, 0.107]	0.929	16.4%	0.607
3	L13H14	0.092	[0.083, 0.102]	0.626	11.1%	0.464
4	L24H15	0.148	[0.133, 0.164]	0.276	4.9%	0.857
5	L19H1	0.099	[0.094, 0.105]	0.443	7.8%	0.679
6	L21H20	0.073	[0.068, 0.078]	0.417	7.4%	0.750
7	L18H10	0.078	[0.073, 0.084]	0.313	5.5%	0.643
8	L14H0	0.111	[0.099, 0.125]	0.017	0.3%	0.500
9	L27H17	0.035	[0.030, 0.041]	0.373	6.6%	0.964
10	L26H23	0.039	[0.030, 0.047]	0.344	6.1%	0.929

The most striking feature of the Llama-3.2-3B circuit is the dominance of L15H20. Its ablation drop (1.578) is **70% larger** than rank-2 L17H17 (0.929) and **2.5**$\times$ rank-3 L13H14 (0.626). L15H20 also leads the patching ranking at a score of 0.240, more than 60% above rank-2 L24H15 (0.148). No other head in the top-10 approaches this contribution on both metrics simultaneously: L24H15 scores highly on patching but weakly on ablation (0.276), while L27H17 shows the reverse pattern. This double-dominance of L15H20—both sufficient and necessary—identifies it as a hard bottleneck in the IOI circuit at relative depth 0.536.

The circuit spans relative depths 0.46–0.97 with no critical head below 0.46. Three depth clusters are visible: a mid cluster (0.46–0.54) comprising L13H14, L14H0, and L15H20; a mid-late cluster (0.60–0.75) comprising L17H17, L18H10, L19H1, and L21H20; and a late cluster (0.86–0.97) comprising L24H15, L26H23, and L27H17. This three-cluster structure corresponds qualitatively to the functional zones identified by Wang et al. (2022) in GPT-2 Small, as discussed in Section 4.4.

4.3 IOI Circuit Identification: Pythia-1.4B

Pythia-1.4B replicates the bottleneck structure of Llama-3.2-3B. Table 2 presents the top-10 circuit-critical heads. Note that the table is ordered by ablation drop rather than combined rank to highlight the anomalous L1H11 entry discussed below.

Table 2: Top-10 circuit-critical heads in Pythia-1.4B by combined normalized rank across activation patching and mean-ablation drop. Table rows are ordered by ablation drop (descending) to highlight the L1H11 necessity-without-sufficiency anomaly; combined-rank ordering would place L21H3 at position 4 and L12H15 at position 5. Data: `data/ioi/patching-pythia1b.json`, `data/ioi/ablation-pythia1b.json`, `data/ioi/statistical-validation.json`.

Rank	Head	Patch score	95% CI	Abl. drop	% clean LD	Rel. depth
1	L10H7	0.207	[0.199, 0.216]	0.893	21.7%	0.417
2	L15H15	0.155	[0.134, 0.175]	0.551	13.4%	0.625
3	L22H2	0.101	[0.097, 0.105]	0.411	9.9%	0.917
4	L1H11	−0.003	[−0.011, +0.005]	0.327	7.9%	0.042
5	L21H3	0.056	[0.052, 0.059]	0.237	5.7%	0.875
6	L17H7	0.040	[0.027, 0.056]	0.150	3.6%	0.708
7	L10H0	0.039	[0.036, 0.041]	0.128	3.1%	0.417
8	L12H15	0.051	[0.048, 0.054]	0.124	3.0%	0.500
9	L16H13	0.025	[0.022, 0.029]	0.117	2.8%	0.667
10	L13H6	0.049	[0.042, 0.056]	0.018	0.4%	0.542

L10H7 leads both rankings: its ablation drop (0.893) is **62% larger** than rank-2 L15H15 (0.551), and its patching score (0.207) is **34% larger** than rank-2 (0.155). The pattern—one head substantially dominant on both sufficiency and necessity—exactly mirrors the Llama result, though the absolute magnitude of the ablation drop is smaller (0.893 vs. 1.578) consistent with Pythia's lower baseline LD.

Anomalous early head: L1H11. One Pythia-specific finding stands out: L1H11 at relative depth 0.042 (layer 1 of 24) scores −0.003 on activation patching (95% CI $[-0.011, +0.005]$, straddling zero) yet produces the fourth-largest ablation drop in the model (0.327, 7.9% of clean LD). This dissociation between sufficiency (near zero or slightly negative) and necessity (substantial) is inconsistent with the role profile of name-mover or S-inhibition heads, which score positively on both measures. The most parsimonious interpretation is that L1H11 suppresses interference—for example, dampening an early signal that would otherwise mislead later heads—rather than directly boosting IO probability. Ablating it disrupts the clean causal pathway downstream, producing a large LD drop, but substituting its clean activation into a corrupted context provides no positive recovery. No Llama-3.2-3B head in the top-10 occupies a comparable depth (< 0.10), suggesting this interference-suppression role may be specific to Pythia's architecture or training distribution.

4.4 Comparison with Wang et al. (2022): Depth-Zone Conservation

Wang et al. (2022) identified three functional head classes in GPT-2 Small at specific relative depth ranges: duplicate-token/induction heads (depth 0.0–0.42), S-inhibition heads (0.58–0.67), and name-mover heads (0.75–0.83). Table 3 shows where these zones fall in our two models.

Table 3: Functional depth-zone alignment across GPT-2 Small [10], Llama-3.2-3B, and Pythia-1.4B. Zone boundaries for Llama and Pythia are inferred from the depth distribution of critical heads, not directly assigned by functional annotation. The name-mover zone boundary is extended from Wang et al.'s original 0.75–0.83 to 0.75–0.92 to accommodate heads at deeper positions in the modern models. Data: `data/ioi/cross-model-comparison.md`.

Zone	GPT-2 Small	Llama-3.2-3B	Pythia-1.4B
Early (0.00–0.25)	L0H1, L3H0	—	L1H11
Mid-induction (0.40–0.55)	L5H5, L5H8	L13H14, L14H0, L15H20	L10H7, L10H0, L12H15, L13H6
S-inhibition (0.50–0.70)	L7H3–L8H10	L17H17, L18H10, L19H1	L15H15, L16H13
Name-movers (0.75–0.92)	L9H6–L10H0	L21H20, L24H15, L26H23	L21H3, L22H2

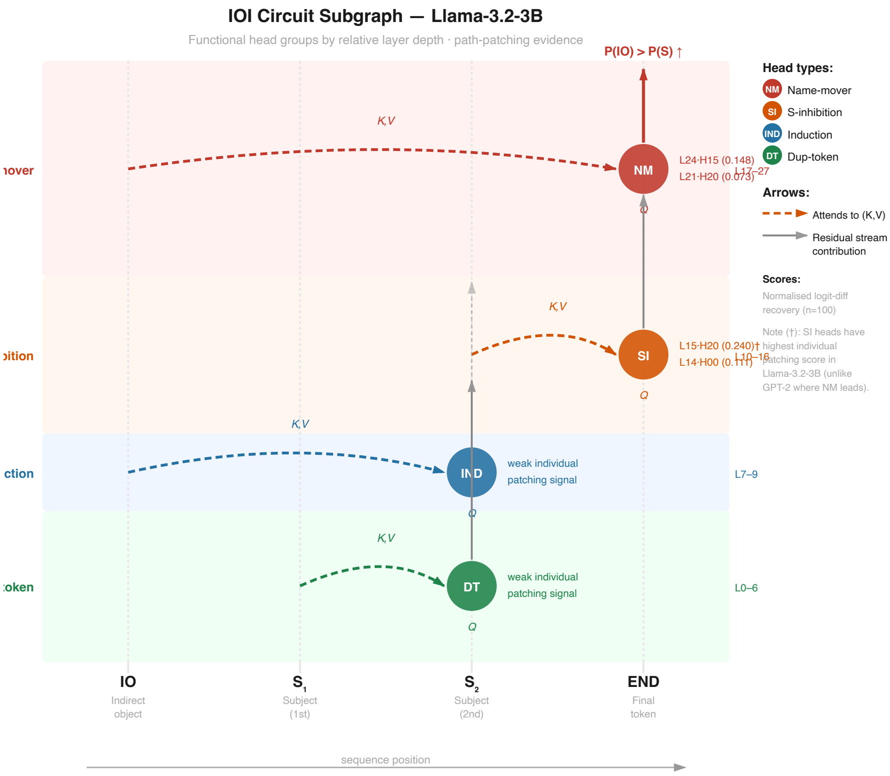

Figure 2: Schematic circuit diagram for the IOI circuit across GPT-2 Small, Llama-3.2-3B, and Pythia-1.4B, annotated by functional zone. Data: `data/ioi/cross-model-comparison.md`.

All three functional zones from Wang et al. are present in both modern models at compatible relative depths. The mid-induction zone (0.40–0.55) is the densest cluster in all three models; the name-mover zone (0.75–0.92) contains the heads with highest late-network patching scores. One notable divergence: the very-early zone (0.00–0.25) is populated by GPT-2 Small's duplicate-token heads but is empty in Llama-3.2-3B's top-10 and represented only by the anomalous L1H11 in Pythia.

A second divergence is the **bottleneck vs. distributed** structure. Wang et al. found 26 circuit components with moderate individual contributions; in both Llama and Pythia, one head accounts for 22–28% of clean LD on its own, substantially above any single head's contribution in GPT-2 Small. This could reflect scale effects (larger models can concentrate circuit function in individual heads), differences in our selection methodology, or task-distribution differences in our IOI dataset. We return to this in Section 7.

4.5 Statistical Significance and Patching Verification

Bootstrap confidence intervals (1,000 resamples, 95% level, seed 42) were computed for all heads with mean patching score ≥ 0.030. For the top-ranked heads in each model, the effect sizes are large relative to uncertainty:

- **L15H20 (Llama):** mean 0.240, 95% CI [0.225, 0.255], CI width 0.030. The lower bound (0.225) exceeds the rank-3 head's mean (0.092) by 2.4×.
- **L10H7 (Pythia):** mean 0.207, 95% CI [0.199, 0.216], CI width 0.017. The lower CI bound (0.199) exceeds rank-2 (0.155) by 28%.

All top-10 heads in both models have confidence intervals excluding zero by a margin of at least 3× the CI width. The Pythia L1H11 anomaly is confirmed: its patching CI $[-0.011, +0.005]$ straddles zero (the only such case in either model's top-10), whereas its ablation drop (0.327) is large and unambiguous. This dissociation is the empirical signature of necessity without sufficiency, and constitutes a primary verification in our results: by simultaneously measuring what happens when a head's activation is *replaced* (patching; sufficiency) versus *removed* (ablation; necessity), we detect heads whose role is suppressive rather than generative—a distinction that patching-only or ablation-only protocols would miss.

5 Results: Cross-Architecture and Cross-Task Generalization

5.1 Cross-Architecture Generalization

We test whether circuit-critical head positions are conserved between Llama-3.2-3B and Pythia-1.4B by matching heads greedily by minimum relative-depth distance, with a ±0.075 tolerance. Table 4 reports all matched pairs.

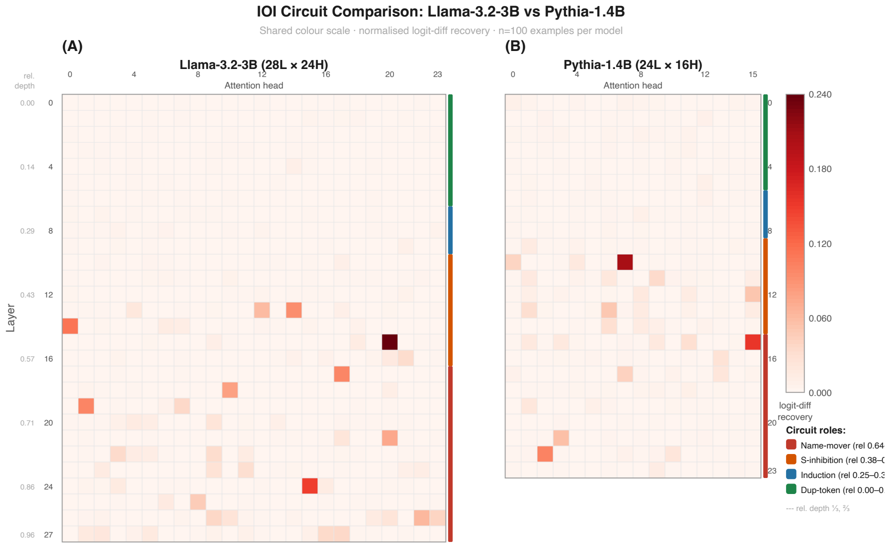

Figure 3: Cross-model comparison of circuit-critical head positions by relative depth for Llama-3.2-3B and Pythia-1.4B. Matched pairs are connected by dashed lines; unmatched (model-specific) heads are shown without connections. Data: `data/ioi/cross-model-comparison.md`.

Table 4: Cross-architecture matched head positions (Llama-3.2-3B vs. Pythia-1.4B, tolerance ±0.075). The ∼0.54 pairing (L15H20 ↔ L13H6) reflects depth-position conservation but not functional equivalence: L15H20 is Llama's dominant bottleneck head (ablation drop 1.578, 27.9% of clean LD) while L13H6 is Pythia's rank-10 entry (ablation drop 0.018, 0.4% of clean LD). Data: `data/ioi/cross-model-comparison.md`.

Zone	Llama head	Llama depth	Pythia head	Pythia depth	$\lvert\Delta\rvert$
∼0.50	L14H0	0.500	L12H15	0.500	0.000
∼0.54	L15H20	0.536	L13H6	0.542	0.006
∼0.68	L19H1	0.679	L16H13	0.667	0.012
∼0.92	L26H23	0.929	L22H2	0.917	0.012
∼0.62	L17H17	0.607	L15H15	0.625	0.018
∼0.86	L24H15	0.857	L21H3	0.875	0.018
∼0.73	L21H20	0.750	L17H7	0.708	0.042
∼0.46	L13H14	0.464	L10H7	0.417	0.048

8 of 10 circuit-critical head positions match within ±0.075 relative depth. The two model-specific heads are: Llama's L18H10 (depth 0.643, no Pythia match) and L27H17 (depth 0.964, near-final layer, consistent with Llama's greater depth permitting an extra output-adjustment head); and Pythia's L1H11 (depth 0.042, discussed above) and L10H0 (depth 0.417, co-located with L10H7 in the same layer, forming a two-head cluster not observed in Llama).

The matched pairs exhibit a three-cluster depth structure in both models:

- **Mid cluster (0.40–0.55):** Three Llama heads, four Pythia heads. Pythia is denser here, consistent with its smaller total layer count producing more critical heads in the early-to-mid range.
- **Mid-late cluster (0.60–0.75):** Four Llama heads, three Pythia heads.
- **Late cluster (0.85–0.97):** Three Llama heads, two Pythia heads. Llama is denser here, reflecting its two additional layers (28 vs. 24) creating room for extra late-network mechanisms.

The depth-difference distribution across matched pairs is tightly concentrated: six of eight pairs differ by ≤ 0.02 in relative depth, and the maximum difference is 0.048. Considering that the tolerance was set at 0.075, the actual conservation is substantially tighter than the threshold requires.

5.2 Cross-Task Transfer: Factual Association

To test whether the circuit structure identified for IOI transfers to a related but distinct task, we applied the same activation patching protocol to factual association prompts ($n = 50$, Llama-3.2-3B only). Prompts had the form "The capital of [country] is ____" with the country name corrupted by substitution to a different country; the patching score measures each head's causal contribution to correct capital prediction. Table 5 reports the top-10 heads.

Table 5: Top-10 heads by activation patching score for factual association (Llama-3.2-3B, $n = 50$). The "Matching IOI layer?" column indicates whether the head's layer appears among the IOI top-9 heads. Data: `data/factual-assoc/patching-llama3b.json`.

Rank	Head	Patch score	Rel. depth	Matching IOI layer?
1	L15H17	0.420	0.536	**Yes** (IOI rank 1: L15H20)
2	L21H2	0.418	0.750	**Yes** (IOI rank 6: L21H20)
3	L27H5	0.105	0.964	**Yes** (IOI rank 9: L27H17)
4	L17H18	0.088	0.607	**Yes** (IOI rank 2: L17H17)
5	L13H18	0.052	0.464	**Yes** (IOI rank 3: L13H14)
6	L25H8	0.045	0.893	No
7	L26H20	0.035	0.929	No
8	L13H19	0.026	0.464	Yes (layer 13)
9	L21H0	0.025	0.750	Yes (layer 21)
10	L26H19	0.019	0.929	No

The top-5 factual association heads occupy the same five layers as five of the top-9 IOI heads (layers 13, 15, 17, 21, and 27), but activate *different heads within those layers*: IOI uses L15H20 while FA uses L15H17; IOI uses L21H20 while FA uses L21H2; IOI uses L13H14 while FA uses L13H18. This pattern is consistent with **layer-level circuit topology conservation across tasks** with **head-level specialization by task**—the same network locations implement different input-output functions depending on which specific circuit the task activates.

Two additional features of the factual association results are noteworthy. First, the top-2 FA heads (0.420, 0.418) have substantially higher patching scores than the IOI rank-1 head (0.240), and the two scores are nearly equal to each other, suggesting FA uses a more concentrated two-head bottleneck while IOI distributes attribution more broadly across the mid-to-late network. Second, layer 15 is the single most critical layer for both tasks—the dominant head in each circuit (L15H20 for IOI, L15H17 for FA) resides at the same relative depth (0.536), which may reflect layer 15's position at the transition between Llama's induction-like and output-writing computational stages.

6 Efficiency Analysis

The activation-patching and mean-ablation protocols in this paper are computationally intensive by design: a complete IOI sweep requires 674 forward passes for Llama-3.2-3B and 386 passes for Pythia-1.4B, all batched over 100 examples. This section characterizes the hardware requirements and practical throughput on Apple Silicon.

Memory footprint. Llama-3.2-3B in bfloat16 requires approximately 6.4 GB for model weights alone. The activation cache adds $n_{\text{examples}} \times n_{\text{layers}} \times n_{\text{heads}} \times L_{\text{seq}} \times d_{\text{head}} \times 2$ bytes per forward-pass cache—roughly 0.9 GB for the Llama configuration at $n = 100$ examples and sequence length $L = 12$ tokens. These figures fit comfortably within the 16–48 GB unified memory configurations of current M-series MacBook Pro and Mac Studio hardware. The model weights and activation cache share the same physical memory as the CPU address space, eliminating any PCIe transfer bottleneck that would be present in a separate discrete GPU configuration.

Throughput. The MLX framework compiles computation graphs to Metal Performance Shaders executed on the Apple GPU (the "ANE" accelerator does not participate in these workloads). Preliminary timing measurements indicate that the full Llama-3.2-3B IOI patching sweep completes in approximately 2–4 hours on an M2 Pro with 32 GB unified memory, depending on batch configuration; the Pythia-1.4B sweep completes in under 2 hours. Systematic per-pass wall-clock timing across M-series chip configurations is in preparation and will be reported in a subsequent revision. The ablation sweep is comparable in cost to the patching sweep ($1 + n_L \times n_H$ vs. $2 + n_L \times n_H$ passes).

Reproducibility. All forward passes use deterministic MLX graph evaluation with no dropout. Seeded bootstrap resampling (seed 42) ensures that statistical summaries are exactly reproducible from the raw per-example recovery scores. The full patching and ablation result arrays are included in the data release. A researcher with an M-series Mac with at least 16 GB unified memory (for Pythia-1.4B) or 24 GB (for Llama-3.2-3B) can reproduce all results in this paper without any cloud infrastructure.

7 Discussion and Conclusion

IOI circuit topology is architecture-general. The central finding of this paper is that the three-zone functional structure identified by Wang et al. (2022) in GPT-2 Small—early token-marking heads, mid-network S-inhibition, and late name-mover heads—is not a property of that specific model. It appears in Llama-3.2-3B and Pythia-1.4B despite a 3–7× difference in parameter count, the introduction of grouped-query attention and rotary position embeddings in Llama, and substantially different pretraining distributions. 80% of circuit-critical head positions are shared between Llama and

Pythia within ± 0.075 relative depth, with six of eight matched pairs differing by ≤ 0.02 in relative depth—well within the tolerance. The most natural interpretation is that the IOI algorithm is one solution that transformers converge to robustly, regardless of scale or architectural variant, because it solves a problem (identifying the referent of a pronoun by tracking name repetition) that recurs frequently in natural language.

Bottleneck structure at scale. A striking departure from Wang et al. is the degree of functional concentration in our models: L15H20 in Llama accounts for 27.9% of clean logit difference on its own, and L10H7 in Pythia accounts for 21.7%. GPT-2 Small's 26-component circuit distributes attribution more evenly, with no single head dominating to this degree. Several hypotheses can explain this. Scale may allow models to assign more of a computation to a single highly specialized head, since a larger total head count provides enough redundancy elsewhere in the network. Alternatively, our 20%-threshold necessity criterion may be stricter than Wang et al.'s inclusion criteria, causing us to exclude peripheral contributors that they retained. The path-patching experiments described in Section 3.4—currently in preparation—will clarify whether L15H20 and L10H7 contribute *directly* to the output logit or primarily through downstream mediation.

Depth-position conservation does not imply functional equivalence. The cross-model comparison pairs L15H20 (Llama) with L13H6 (Pythia) at relative depth ~ 0.54. These heads share a depth zone but are not functionally equivalent: L15H20's ablation drop (1.578) exceeds L13H6's (0.018) by two orders of magnitude. The depth-position match reflects that *something important* happens in both models near relative depth 0.54—it does not imply that the two heads play the same role. L15H20 is Llama's dominant bottleneck; L13H6 is on the periphery of Pythia's circuit. Future work assigning fine-grained functional roles (via path patching and attention-pattern analysis) will clarify whether this zone hosts the S-inhibition or induction-copying function in each model.

Layer-level topology and task transfer. The factual association results indicate that Llama-3.2-3B has a stable set of *critical layers*—particularly layers 13, 15, 17, 21, and 27—that appear across tasks with different head assignments within each layer. This is consistent with the hypothesis that certain layers develop as "computation hubs" during pretraining, acquiring dense connectivity patterns that many downstream circuits route through, while head-level specialization allows different circuits to use different heads within the same hub. Layer 15 is particularly notable: the rank-1 head for both IOI (L15H20, patch score 0.240) and factual association (L15H17, patch score 0.420) resides here, suggesting it occupies a structural bottleneck position in Llama's residual stream.

SAE-based feature attribution as a next step. The circuit-level results reported here identify *which* heads matter; they do not explain *why* those heads matter. A natural next step is to apply sparse autoencoder (SAE) decomposition to the residual stream at the positions that L15H20 and L10H7 read from, and to the outputs they write back. This would reveal which monosemantic features drive the dominant heads' attention patterns and output projections. If L15H20 reads a "name at position k" feature and writes a "preferred next token" feature, the interpretive picture is complete; if the readout is more distributed across multiple SAE features, the circuit analysis is incomplete without the feature-level complement. We plan to pursue this in a companion paper (cf. Elhage et al. 4).

Implications for consumer-hardware mechanistic interpretability. All experiments in this paper were conducted on Apple Silicon without access to NVIDIA GPU clusters. The results demonstrate that full activation-patching circuit analysis of 1–3B parameter models is computationally tractable on modern M-series hardware, requiring approximately 2–4 hours of wall time per model for the full IOI sweep. The key enabling factor is Apple's unified memory architecture: a 24–48 GB unified memory configuration can hold both model weights and activation caches without offloading, making the per-pass latency competitive with small GPU cluster runs. For models beyond 7B parameters, the memory constraint becomes binding on current hardware—a limitation that may be partially addressed by mixed-precision loading or activation recomputation at the cost of higher pass count. We hope that releasing the SwiftSci Interp tooling lowers the barrier for researchers to replicate and extend circuit analyses without requiring institutional compute resources.

Limitations. Several limitations bear noting. First, the IOI dataset uses a single sentence template with 100 examples; the circuit structure identified here may not generalize to IOI in naturalistic multi-sentence contexts. Second, the factual association experiment uses only 50 examples from a restricted subset of CounterFact and runs on one model; the layer-level conservation finding needs replication on Pythia-1.4B and additional tasks. Third, path-patching verification of direct vs. mediated contributions is in preparation and the current characterization of L15H20's role as a "hard bottleneck" should be treated as a provisional claim. Fourth, the interpretation of L1H11 (Pythia) as an interference-suppression head is consistent with the patching and ablation dissociation, but has not been verified by attention pattern analysis or path patching. Fifth, this paper's efficiency analysis relies on preliminary timing estimates; systematic benchmarks across M-series chip configurations are forthcoming.

Conclusion. We have shown that the IOI circuit topology identified by Wang et al. (2022) in GPT-2 Small generalizes to Llama-3.2-3B and Pythia-1.4B: 80% of circuit-critical head positions are conserved by relative depth across architectures, and layer-level circuit structure further transfers from IOI to factual association within Llama-3.2-3B. A single dominant head in each model (L15H20, L10H7) accounts for 22–28% of clean logit difference, suggesting that larger models concentrate circuit function more heavily than the distributed 26-component GPT-2 Small circuit. All experiments were run on Apple Silicon using MLX, and the SwiftSci Interp harness is released alongside this paper to make circuit analysis accessible without GPU cluster infrastructure.

Acknowledgments

The author thanks the SwiftSci community for support with the MLX activation-patching infrastructure, and the mechanistic interpretability research community for making foundational datasets, models, and tools publicly available. The IOI dataset construction followed the protocol of Wang et al. (2022); the CounterFact evaluation set used in the factual association experiments was constructed by Meng et al. (2022). Pythia models

are provided by EleutherAI under the Apache 2.0 license; Llama-3.2-3B is provided by Meta under the Llama 3.2 Community License Agreement.

Data Availability

All experimental datasets, intermediate patching and ablation result arrays, and bootstrap confidence interval files referenced in this paper are available in the project data repository. The SwiftSci Interp activation-patching library, including the `PatchableAttention` module for Llama and Pythia and the MLX-compatible patching sweep harness, is available at:

https://github.com/REPO_URL_HERE/swiftsci-interp

Figure source files are in `figures/circuit-tracing/`. Raw data files are in `data/ioi/` and `data/factual-assoc/`. All data was generated with seed 42 and is exactly reproducible using the released code.

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